BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2025

MATHEMATICS

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions)

Maximum Marks: 40

Note: The section header says (40) but the note below it says 20 part questions; the scan shows 20 MCQs (i–xx), all transcribed. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 100 (Science Pre-Engineering Group)

Q1. Choose the correct answer for each from the given options:

Qi. If $Z = -3i + 4$, then $\bar{Z}$ =:

(A) $3i + 4$
(B) $3i - 4$
(C) $-3i + 4$
(D) $-3i - 4$

Qii. For n is an even integer, then i^n is equal to:

(A) i
(B) $-i$
(C) 1 or -1
(D) i or -1

Qiii. The number of non-zero rows in echelon from of a matrix is called:

(A) Order of a matrix
(B) Rank of matrix
(C) Leading column
(D) Leading row

Qiv. The cofactor A_{22} of $[[1,2,4],[-1,2,5],[0,1,-1]]$ =:

(A) 0
(B) -1
(C) 1
(D) 2

Qv. If O be the urges and $\vec{OP} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{OQ} = 5\hat{i} + 4\hat{j} - 3\hat{k}$, then $\vec{PQ}$ is equal to:

(A) $7\hat{i} + 7\hat{j} - 7\hat{k}$
(B) $-3\hat{i} + \hat{j} - \hat{k}$
(C) $3\hat{i} + \hat{j} + \hat{k}$
(D) $-7\hat{i} + 7\hat{j} + 7\hat{k}$

Qvi. The volume of tetrahedron determined by the vectors $\bar{a}$, $\bar{b}$ and $\bar{c}$ is:

(A) $[\bar{a},\bar{b},\bar{c}]$ (B) $1/3[\bar{a},\bar{b},\bar{c}]$

(C) $1/6[\bar{a},\bar{b},\bar{c}]$ (D) $\bar{a}.(\bar{b} \times \bar{c})$

Qvii. If $a_n - a_{n-1} = n + 1$ and $a_n = 14$ then a_5 =:

(A) 3 (B) 5

(C) 14 (D) 20

Qviii. If (a^{n+1}+b^{n+1})/(a^n+b^n) is an A.M between a and b, then n is equal to:

(A) 1 (B) 2

(C) 0 (D) 3

Qix. $\Sigma(n+1 \text{ to } 10)\ n^0$ is:

(A) 10 (B) 9

(C) 12 (D) 8

Qx. If n = 0, then (n+1)!/n! =:

(A) 0 (B) 1

(C) n (D) ∞

Qxi. The number of ways that a machine of n beads of different colours be made is:

(A) (n + 1)! (B) n!/2

(C) (n−1)!/2 (D) (n − 1)!

Qxii. The number of terms in the expansion of $(1 + x)^7$ is:

(A) 7 (B) 8

(C) 9 (D) 10

Qxiii. For |x| < 1 then $1 - x + x^2 - x^3 + \ldots$ is equal to:

(A) $(1 + x)^{-1}$ (B) $(1 - x)^{-1}$

(C) $(1 + x)^{-2}$ (D) $(1 - x)^{-2}$

Qxiv. Inverse of a function exists if f it is:

(A) Injective (B) Bijective

(C) Surjective (D) Inverse

Qxv. Domain of the function $f(x) = x^2/(x^2-4)$ is the set of all real numbers except:

(A) 4 (B) 3

(C) 2 (D) ±2

Qxvi. $\sin(\theta/2)$ is equal to:

(A) $\pm\sqrt{((1+\sin\theta)/2)}$ (B) $\pm\sqrt{((1-\cos\theta)/2)}$

(C) $\pm\sqrt{((1+\cos\theta)/2)}$ (D) $\pm\sqrt{((1-\sin\theta)/2)}$

Qxvii. Hero's formula is:

(A) $\blacktriangle = s(s-a)(s-b)(s-c)$ (B) $\blacktriangle = \sqrt{((s-a)(s-b)(s-c))}$

(C) $\blacktriangle = \sqrt{(s(s-a)(s-b)(s-c))}$ (D) $\blacktriangle = (a+b+c)/2$

Qxviii. A circle which touches one side of the triangle externally and two extended sides internally is called:

(A) circum-circle (B) in-circle

(C) tri-circle (D) escribed-circle

Qxix. The graph of a linear function is:

(A) straight line (B) circle

(C) parabola (D) triangle

Qxx. $\cos(\sin^{-1}(1/2)) =$:

(A) $2/\sqrt{3}$ (B) $2\sqrt{3}$

(C) $\sqrt{3}/2$ (D) $1/(2\sqrt{3})$

Section 'B' (Short-Answer Questions)

Maximum Marks: 40

> Note: Attempt any TEN part questions in all. All questions carry equal marks. (i.e. 4 marks of each part).

Q2.

(i) Separate the real and imaginary parts of $(\sqrt{2} + i)^{-2}$.

(ii) Using properties, Prove that: $|\alpha\ \beta\gamma\ \alpha\beta\gamma;\ \beta\ \gamma\alpha\ \alpha\beta\gamma;\ \gamma\ \alpha\beta\ \alpha\beta\gamma| = |\alpha\ \alpha^2\ \alpha^3;\ \beta\ \beta^2\ \beta^3;\ \gamma\ \gamma^2\ \gamma^3|$

(iii) Show that $[[1,-2,-6],[-3,2,9],[2,0,-3]]$ is a periodic matrix of period 2.

(iv) Find the angle between the vectors $3\hat{i} + 5\hat{j} - 3\hat{k}$ and $\hat{i} - 4\hat{j} + \hat{k}$.

(v) Insert three arithmetic means between 3 and 11.

(vi) In a G.P. $a_3 = 12$ and $a_8 = 384$. Find the nth term.

(vii) Find the sum of the series $\Sigma(k=2 \text{ to } \infty)(1/(k^2-1))$.

(viii) If ${}^nP_3 = 12$, ${}^{n}\!/_{2}P_3$ Find n.

(ix) Prove that by using principle of mathematical induction $(7^n - 4^n)$ divisible by 3.

(ix-OR) Prove that by using principle of mathematical induction. $1^3 + 2^3 + 3^3 + \ldots + n^3 = [(n(n+1))/2]^2$, for all natural numbers n.

(x) Find inverse of f and determine and domain and range f^{-1} for the real valued function $f(x) = (x-1)/(x-3)$, $x \neq 3$.

(xi) Prove that: $(\sin 6\theta + \sin 4\theta)/(\cos 6\theta + \cos 4\theta) = \tan 5\theta$

(xii) Find the smallest angle in ΔABC when a = 6cm, b = 8cm and c = 9.4cm.

(xiii) Find the value of R and r in ΔABC when a = 40cm, b = 12cm and c = 38cm.

Section 'C' (Detailed Answer Questions)

Maximum Marks: 40

Note: Attempt any FIVE question from this section. All questions carry equal marks.

Q3.

Use Gauss-Jordan method to solve the system of linear equation: $x - y + 4z = 4$; $2x + 2y - z = 2$; $3x - 2y + 3z = -3$

Q4.

In a parallelogram ABCD, X is the mid point of AB and Y divides BC in 1:2, Show that if Z divides DX in 6:1, then it also divides AY in 3:4.

Q5.

Two events A and B sure that P(A) = 1/4, P(A/B) = 1/2 and P(B/A) = 2/3. Find $P(A \cap B)$ and P(B).

Q6.

If $x = 1/3 + (1.3)/(3.6) + (1.3.5)/(3.6.9) + \ldots$ then prove that $x^2 + 2x - 2 = 0$.

Q7.

Show that $2\tan^{-1}(2/3) = \cos^{-1}(5/13)$

Q8.

Prove that: $\cos 20° \cos 40° \cos 60° \cos 80° = 1/16$.

Q9.

Find the point of intersection of the functions $y = 2x + 1$ and $y = x^2 - 4x + 6$, $\forall x \in R$ graphically.